Assistive devices for mobility

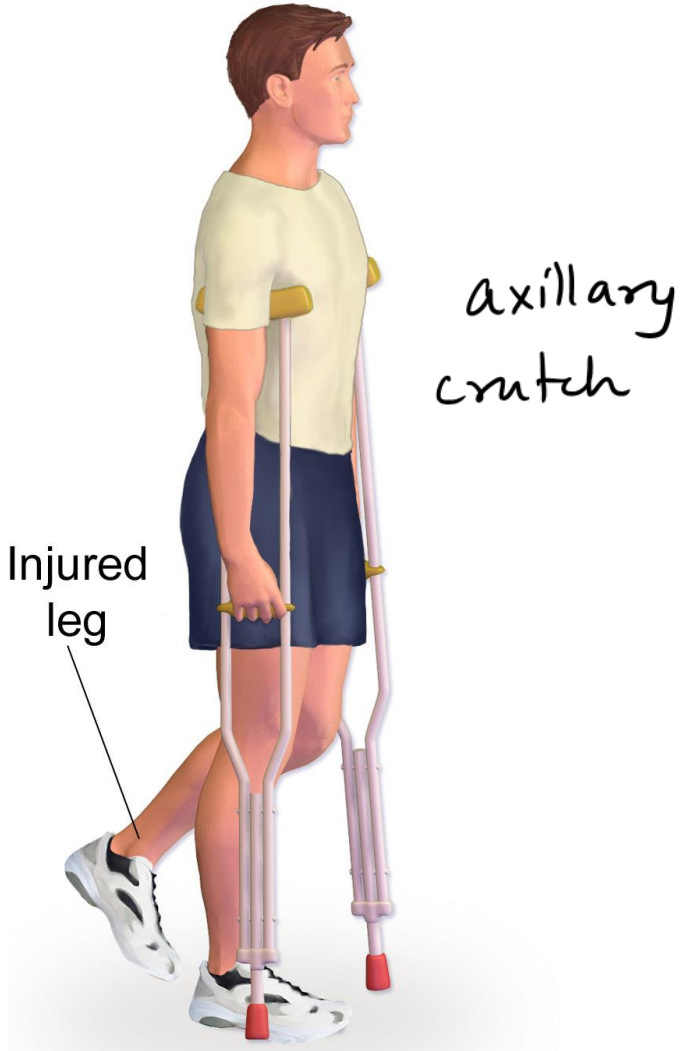


Forearm

Vissco Elbow/Forearm crutch



Ossenberg Elbow/Forearm crutch

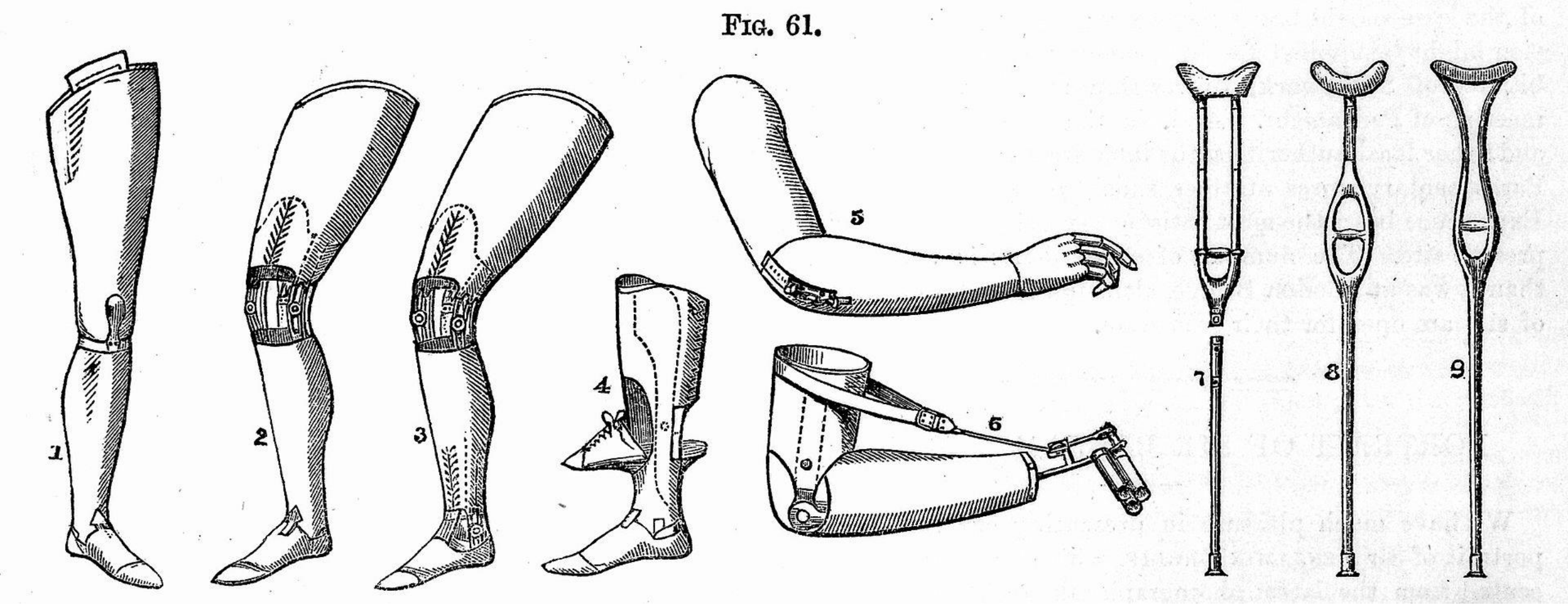


Walking with Crutches

TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

By BruceBlaus - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=59441530

Assistive devices



https://commons.wikimedia.org/wiki/File%3AReport_on_surgical_instruments_at_Int._Exhibition%2C_1862_Wellcome_L0019884.jpg
See page for author [CC BY 4.0 (http://creativecommons.org/licenses/by/4.0)], via Wikimedia Commons

Types of assistive devices for walking

Prostheses — plural

- Prosthesis
  - Replacement for a missing limb
- Orthosis
  - Supportive device to compensate for leg pathologies
- Exoskeleton
  - Device to augment the performance of able-bodied humans

Active device – uses external power sources

Passive device

Semi-active or semi-passive

Lower limb amputation

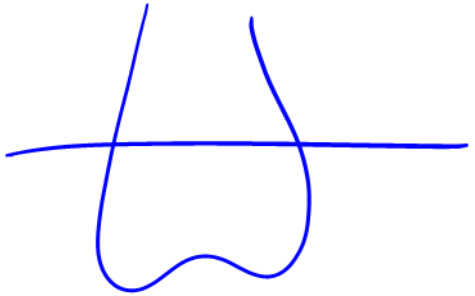
- Vascular disease, mainly caused by diabetes, smoking
- Trauma
  - Accidents, war injury
- Cancerous tumors
- Congenital causes

Oscar Pistorius

Blade Runner

Double amputee

Lower limb amputation



- Partial foot
- Below-knee (B/K) or transtibial
- Above-knee (A/K) or transfemoral
- Hemipelvectomy
- Amputation through the hip, knee or ankle joint

Hip disarticulation

Prosthesis



https://commons.wikimedia.org/wiki/File:Artificial_left_leg_Wellcome_L0037237.jpg



https://www.jaipurfoot.org

Jaipur foot

Prosthesis and Orthosis

Modular design

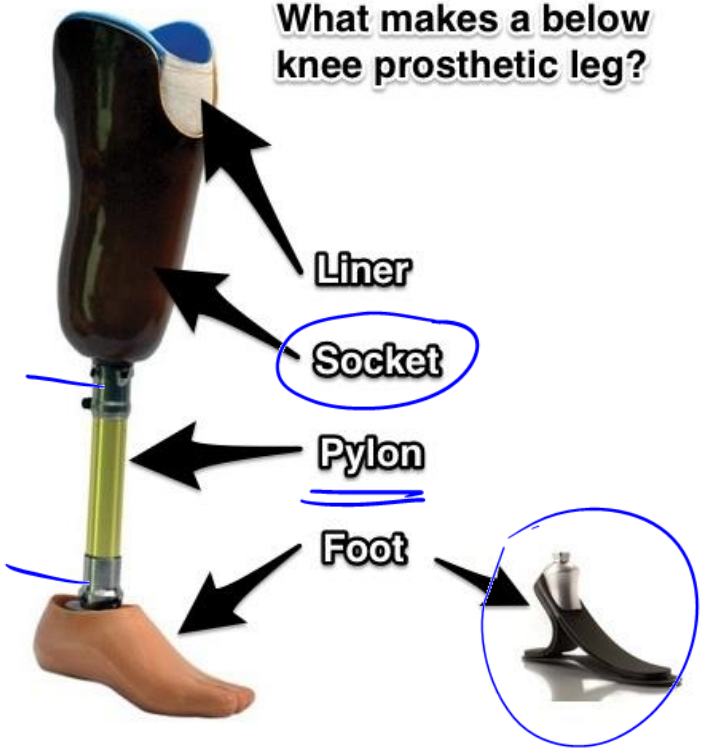


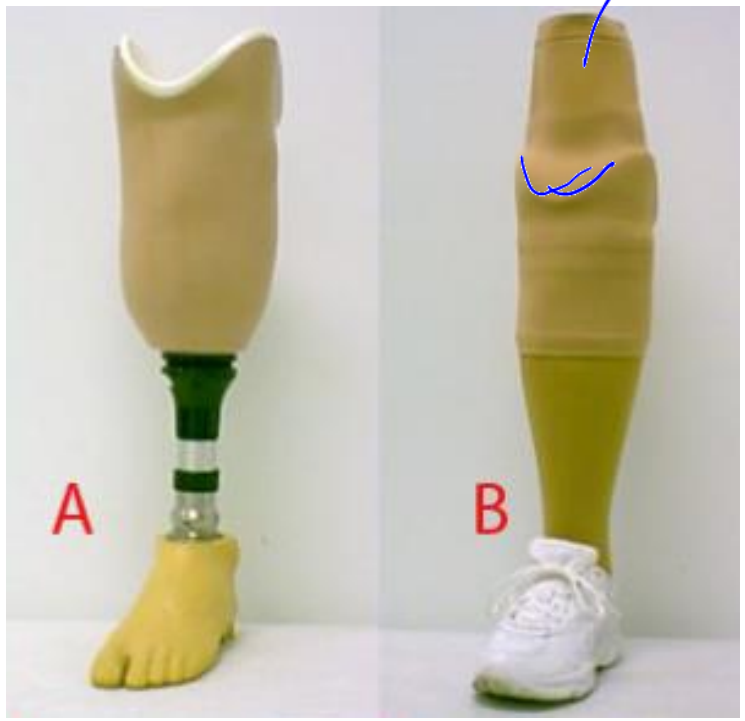


Figure courtesy http://olivercarefull.com/what-makes-a-prosthetic-leg/



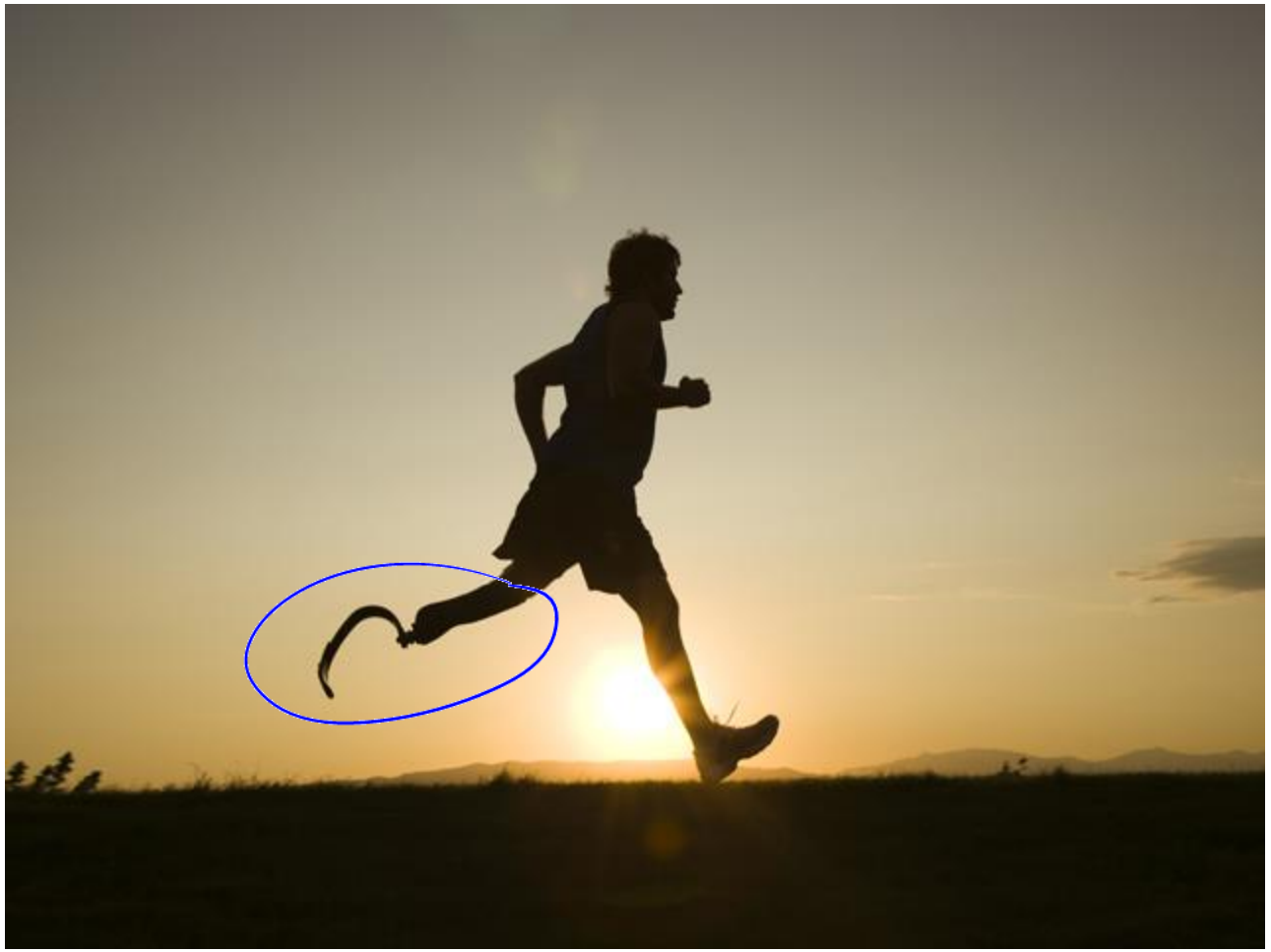
Figure courtesy Dr. Ganesh Bapat, Alumnus R2D2, IITM

Prosthesis



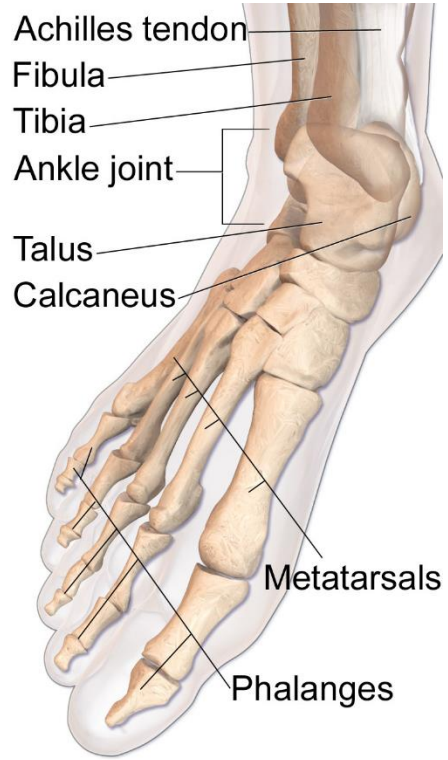
suspension sleeve

http://prosindiana.com/prosthetics/lower-extremity/below-knee/



Human foot

- Mechanical marvel
- 26 bones, 33 joints, and more than 100 tendons, muscles, and ligaments
- One of the most complex parts of the human body
- Multi-functional



Lower Leg and Foot

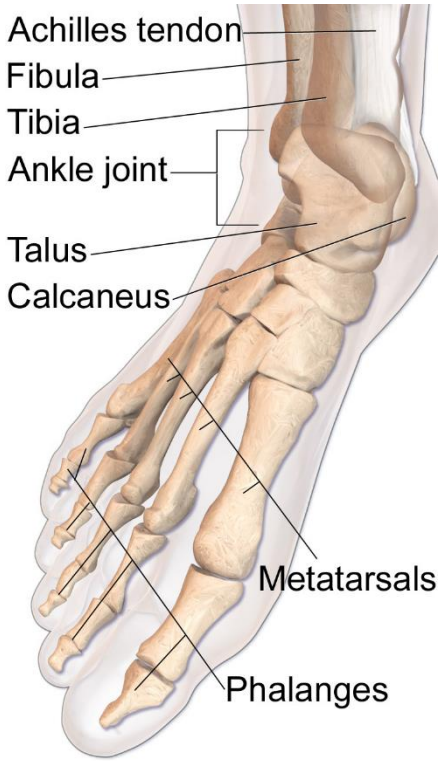
Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29849183

Prosthetic vs Human feet



SACH foot (Otto Bock)

Solid Ankle Cushioned Heel



Lower Leg and Foot

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29849183



http://newsfeed.time.com/2013/02/23/sum-of-the-parts-prosthetics-through-the-centuries/slide/inside-foot/

Jaipur foot

Prosthetic feet



odyssey K3





Magnum® Foot



Carbon Copy 2® Foot